R04: verified progress in the P-256 polynomial checkpoint

Run nist-factor-R04-20261022T0935Z, started 2026-10-22 09:35:14 UTC. One native NFS allocation continues the 444-bit residual of the 512-bit skipped_256_876-twist2 target. The frozen protocol binds the literal input and methods. Exact final counts and execution status are serialized in summary.json; the partition separately records the arithmetic status.

Outcome: exit 137 at the outer bound, no new factor returned. The verified partition remains partial, with a 444-bit composite residual. Final recovery preserves 329 valid candidate records, including repeats, through leading coefficient 900. The production search produced no selected final polynomial or sieve relations. A separate bounded recognition diagnostic then selected a saved polynomial and generated the native job for the next sieving allocation. Both output formats pass exact input, coefficient and common-root checks. This terminated search is inconclusive; the arithmetic total remains eight complete targets and thirteen resolved timeout fields.

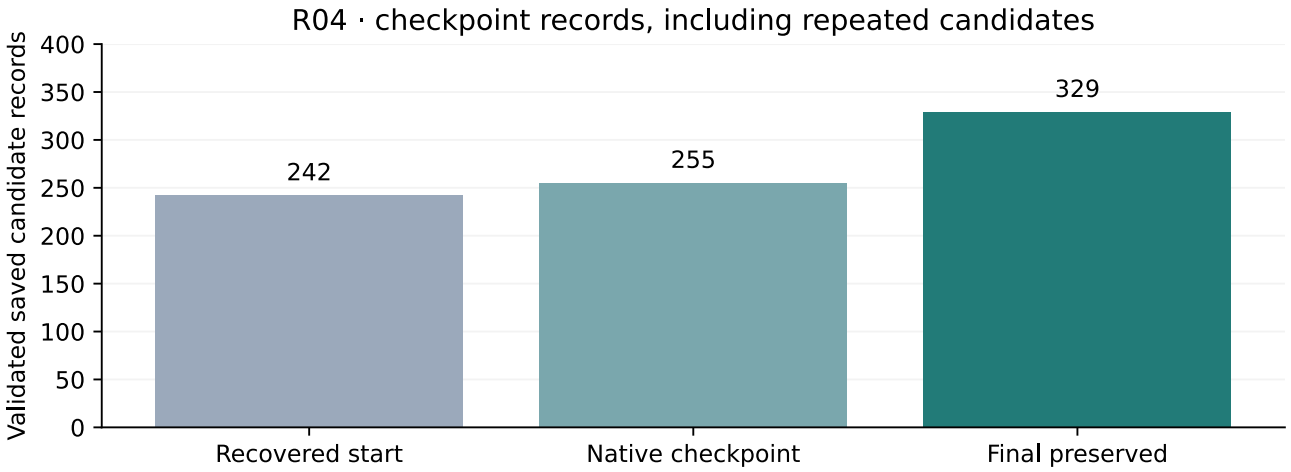
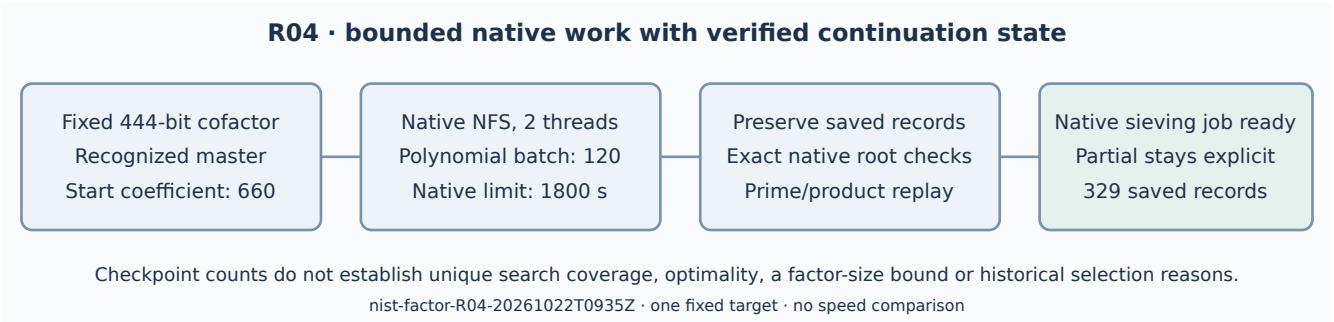


Figure: complete saved candidate records validated by native GP at three fixed checkpoints, one file per stage. Repeats are retained. These are not unique candidate counts or completed coefficient-range counts, and no sampling interval applies. Source fields are linked in plot-data.json. This graph records preservation of usable state, not faster factorization.

Scope and observations

R03 had discovered that a per-thread-only restore silently restarted the polynomial search. Before R04, native exact checks validated 242 saved candidates, wrong-input rejection passed, and a separate diagnostic confirmed native input matching and resume coefficient 660. R04 starts from that recognized master. Its per-thread output begins at 660 and later records pass 800. An intermediate durable native master has 255 verified records, highest coefficient 720, and a verbatim native time: 837 line. This demonstrates actual native checkpoint merging, not just restored bytes.



The method uses unchanged YAFU source c7d90283972820f99b3b4920bfc1eff05c1f6342, -gnfs -xover 81, seed 20261022, two sieve/linear-algebra threads in separate phases, polynomial batch 120, native 1800-second limit, outer 1860 seconds plus 30 seconds kill grace, and 4 GiB virtual-memory limit per process. Per-attempt source commit, executable/siever hashes, command, timestamps, stdout, stderr and receipt are retained. The host has four CPUs and 16 GiB RAM, L0 isolation. No runtime estimate, speed comparison or new factor-size lower bound is inferred from those operational limits.

The legacy receipt's timed_out: false recognizes exit 124 only and does not exclude a grace-kill exit 137. The exact timestamps and status are retained without rewriting it. Future receipt version 2 logs GNU timeout's signals and reports an unclassified exit 137 as unknown, avoiding that ambiguity. Its ignored-SIGINT/grace-kill control is retained separately.

Correctness and durable state

After the native worker stops, complete saved records are preserved and checked for six integer coefficients, a nonzero leading coefficient, invertible Y1 and the exact common-root equation modulo the fixed input. The native master and original thread files remain in raw evidence. The thin shell helper only frames records; all arithmetic is native PARI/GP 2.17.2. Trailing incomplete blocks are retained in originals and excluded from accepted candidate input. Native timing lines are copied verbatim.

The helper rejects a live worker. A real-fixture control, applied twice, keeps the same record count because consumed thread files are retained outside the active state and cannot be merged twice. Wrong-input root rejection is inherited from R03. A copied native master with its actual time record replays byte-for-byte. The new wrapper's prepare-only path restores the fixed input without starting search. Portable checkpoint replay independently checks the whole-target product and every retained prime certificate; a valid partial partition does not imply completion.

The separately preregistered final recognition diagnostic reads coefficient 900 and exits zero at its two-second native limit, inside the three-second outer diagnostic bound. It selects saved record 83, leading coefficient 240, recorded Murphy-E 5.561e-11, and writes the native job and polynomial parameter file. Each file's input and exact common root pass separate GP checks; their coefficients match. The native printed C134 equals the unchanged input and is not a proper factor. The derived job is retained in the final checkpoint so the next allocation starts sieving. Its configured 46,000,000-relation target is not an observed count, runtime estimate or guarantee. No sieve relations were generated by this diagnostic.

The current nine exact product identities and prime certificates remain those of R02 unless the separate result records a new certified split. Lean's scope is product equality, with primality established separately by PARI. The default native identity issuer also rechecks all nine unchanged R01 records, preserving backward compatibility. No library source changed since the 4,479-pass suite; shell checks and backlog validation pass.

Evidence and remaining requirement

The complete round archive and checksum preserve attempts, failures, native states, snapshots, refinement/replay, helper controls and source copies. A separate compact excerpt provides the exact next-job input. The later backlog card is draft and unqueued; no lease, paid task or remote factor worker is created. Python only lays out recorded values and this PDF.

The specific factorization gap still requires a complete prime-certified split of the literal residual; native progress is not an impossibility or security result. R02's separate large-archive publication remains blocked by missing AWS connection setup. Historical generators, upstream trait defects and the historical seed-skip reason remain outside this round's conclusions. Existing IC scoreboards, timelines and performance figures have no affected result and remain unchanged.